Vibrations Workshop 5 - The Electrical-Mechanical Identity Crisis

By: Jake Peyser
ME301
2/25/2026

General Overview

In this assignment, the objective is draw parallels between electrical and mechanical systems by analyzing Newton's Second Law and Kirchhoff's Voltage Law in the Laplace domain.

In particular, for any provided mechanical system, an equivalent circuit diagram will be illustrated. Likewise, for any provided electrical system, an equivalent mechanical diagram will be illustrated. The respective equations of motion will be determined as well in the Laplace domain.

If not clear, both Newton's Second Law and Kirchhoff's Voltage Law will be assumed throughout this analysis with the understanding that:

- Positively Oriented Force $F \longleftrightarrow$ Positively Oriented Voltage Source with Potential Difference F
- Spring with Stiffness $k \longleftrightarrow$ Capacitor with Capacitance $1/k$
- Mass $m \longleftrightarrow$ Inductor with Inductance m
- Damper with Damping Constant $R \longleftrightarrow$ Resistor with Resistance R

An important note is that all diagrams to follow were derived from the provided (illustrated) systems in the separate workshop pdf for this assignment.

Problem (a)

The circuit diagram here was derived from the provided illustrated mechanical system in accordance with the physics. Note that v corresponds with the (relative) coordinate velocities of the respective components.

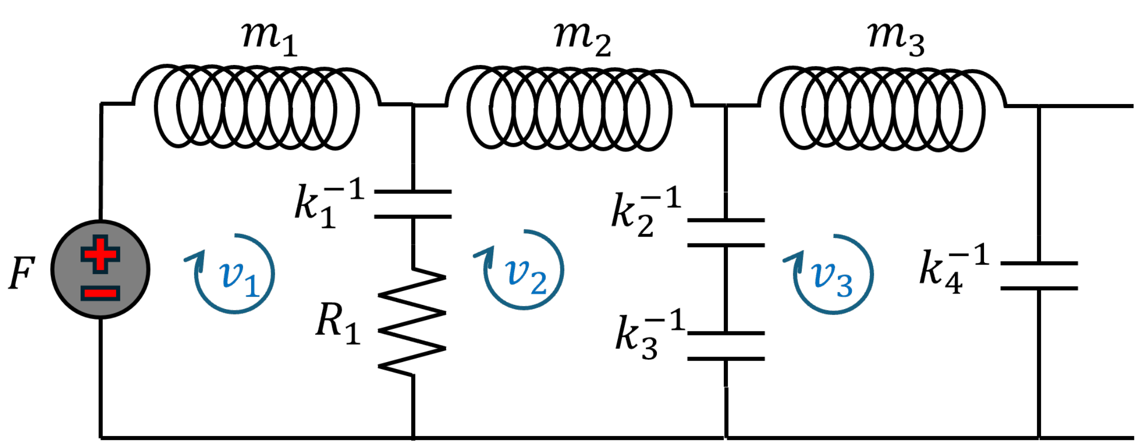

Figure 1: Equivalent Circuit (a)

The equations of motion encoded by this circuit directly follow from Kirchhoff's Voltage Law via impedance (with Laplace parameter s and $\mathscr{L}\{v_i\} = V_i$ for $i = 1, 2, 3$):

$$\begin{bmatrix} m_1 s + \frac{k_1}{s} + R_1 & -\frac{k_1}{s} - R_1 & 0 \\ -\frac{k_1}{s} - R_1 & R_1 + \frac{k_1}{s} + m_2 s + \frac{k_2}{s} + \frac{k_3}{s} & -\frac{k_2}{s} - \frac{k_3}{s} \\ 0 & -\frac{k_2}{s} - \frac{k_3}{s} & \frac{k_4}{s} + \frac{k_3}{s} + \frac{k_2}{s} + m_3 s \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} F \\ 0 \\ 0 \end{bmatrix}$$

Problem (b)

For this problem, the circuit diagram was already provided. To motivate the illustration of the equivalent mechanical system, a coordinate system will be imposed inside the circuit as follows:

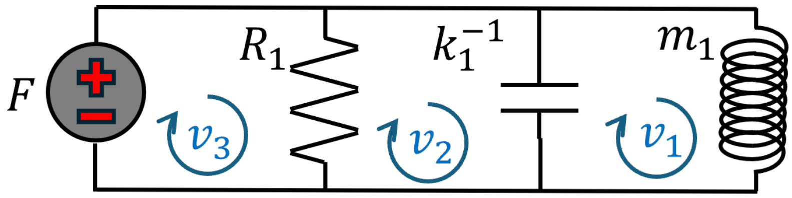

Figure 2: Labeled Circuit (b)

The equations of motion encoded by this circuit directly follow from Kirchhoff's Voltage Law via impedance (with Laplace parameter s and $\mathscr{L}\{v_i\} = V_i$ for $i = 1, 2, 3$):

$$\begin{bmatrix} R_1 & -R_1 & 0 \\ -R_1 & R_1 + \frac{k_1}{s} & -\frac{k_1}{s} \\ 0 & -\frac{k_1}{s} & \frac{k_1}{s} + m_1 s \end{bmatrix} \begin{bmatrix} V_3 \\ V_2 \\ V_1 \end{bmatrix} = \begin{bmatrix} F \\ 0 \\ 0 \end{bmatrix}$$

The equivalent mechanical system follows and is illustrated below in the mechanical diagram with v representing the respective (relative) component coordinate velocity:

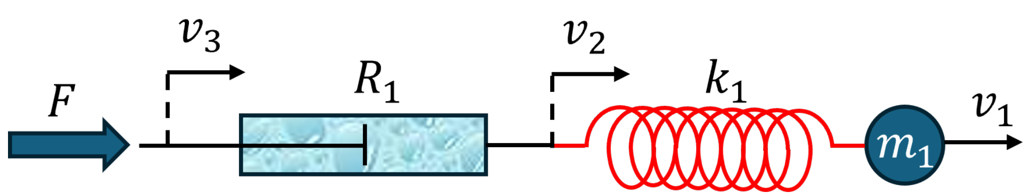

Figure 3: Equivalent Mechanical System (b)

Problem (c)

The circuit diagram here was derived from the provided illustrated mechanical system in accordance with the physics. Note that v corresponds with the (relative) coordinate velocities of the respective components.

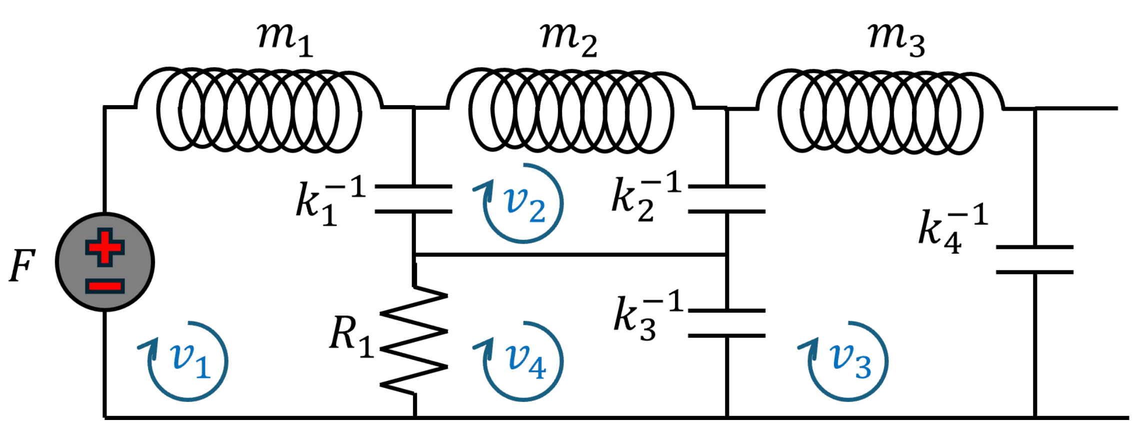

Figure 4: Equivalent Circuit (c)

The equations of motion encoded by this circuit directly follow from Kirchhoff's Voltage Law via impedance (with Laplace parameter s and $\mathscr{L}\{v_i\} = V_i$ for $i = 1, 2, 3, 4$):

$$\begin{bmatrix} m_1 s + \frac{k_1}{s} + R_1 & -\frac{k_1}{s} & 0 & -R_1 \\ -\frac{k_1}{s} & \frac{k_1}{s} + m_2 s + \frac{k_2}{s} & -\frac{k_2}{s} & 0 \\ 0 & -\frac{k_2}{s} & \frac{k_4}{s} + \frac{k_3}{s} + \frac{k_2}{s} + m_3 s & -\frac{k_3}{s} \\ -R_1 & 0 & -\frac{k_3}{s} & R_1 + \frac{k_3}{s} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} F \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Problem (d)

This problem is unlike the others in the sense that the given mechanical system does not correspond with linear motion. To fully grasp the dynamics at play, the system will be labeled as follows:

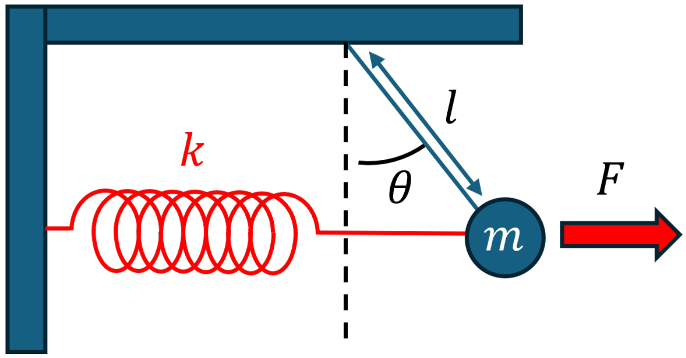

Figure 5: Labeled Mechanical System (d)

The energy breakdown of the system is given as follows (assuming a uniform gravitational field):

- Kinetic: $K = \frac{1}{2}ml^2\dot{\theta}^2$
- Spring Potential: $U_k = \frac{1}{2}k(l\sin\theta)^2$
- Gravitational Potential: $U_g = -mgl\cos\theta$
- Work Done by (constant) Force: $W = \int F d(l\sin\theta) = Fl\sin\theta +$ (negligible constant)
- Net Potential: $V = U_k + U_g - W = \frac{1}{2}k(l\sin\theta)^2 - mgl\cos\theta - Fl\sin\theta$
- Effective Lagrangian: $\mathcal{L} = K - V = \frac{1}{2}ml^2\dot{\theta}^2 - \frac{1}{2}k(l\sin\theta)^2 + mgl\cos\theta + Fl\sin\theta$

It follows in accordance with the small angle approximation that:

- $p_\theta = \frac{\partial\mathcal{L}}{\partial\dot{\theta}} = ml^2\dot{\theta} \implies \dot{p}_\theta = ml^2\ddot{\theta}$
- $Q_\theta = \frac{\partial\mathcal{L}}{\partial\theta} = -kl^2\cos\theta\sin\theta - mgl\sin\theta + Fl\cos\theta \approx -kl^2\theta - mgl\theta + Fl = Fl - (kl^2 + mgl)\theta$

It directly follows from the Euler-Lagrange Equation $\dot{p_\theta} = Q_\theta$, which is equivalent to Newton's Second Law, that the (linearized) equation of motion for the system is:

$$ml^2\ddot{\theta} = Fl - (kl^2 + mgl)\theta \implies ml^2\ddot{\theta} + (kl^2 + mgl)\theta = Fl$$

Upon applying the laplace transform $\mathscr{L} : f \times s \in (C^\infty(\mathbb{R}^+, \mathbb{C}), \mathbb{C}) \mapsto \int_0^\infty e^{-st} f(t)\, dt$ to the above EOM with $\omega = \dot{\theta}$, $\mathscr{L}\{\omega\} = \Omega$, and neglecting all initial conditions, it follows that:

$$\Omega(s) = \frac{Fls}{ml^2s^2 + kl^2 + mgl} \iff \left(ml^2s + \frac{kl^2 + mgl}{s}\right)\Omega(s) = Fl$$

This equation (above), using impedance analysis, is encoded by an RLC (series) circuit with:

- $R = 0 \implies Z_R = 0$
- $L = ml^2 \implies Z_L = ml^2s$
- $C = (kl^2 + mgl)^{-1} \implies Z_C = (kl^2 + mgl)/s$
- $V = Fl \implies Z_V = Fl/\Omega$

These components in series correspond with the following circuit diagram shown right below via Kirchhoff's Voltage Law:

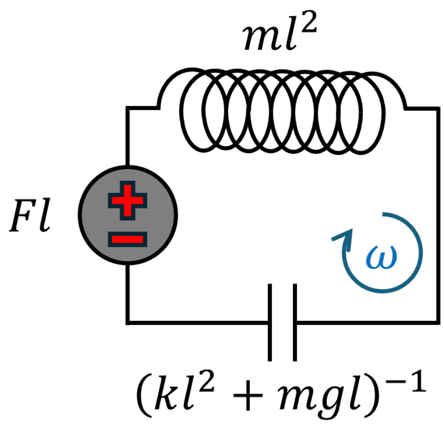

Figure 6: Equivalent Circuit (d)

Problem (e)

The circuit diagram here was derived from the provided illustrated mechanical system in accordance with the physics. Note that v corresponds with the (relative) coordinate velocities of the respective components.

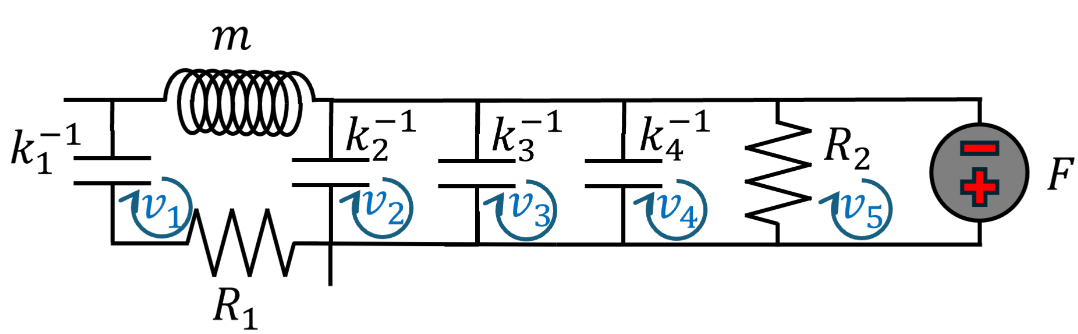

Figure 7: Equivalent Circuit (e)

The equations of motion encoded by this circuit directly follow from Kirchhoff's Voltage Law via impedance (with Laplace parameter s and $\mathscr{L}\{v_i\} = V_i$ for $i = 1, 2, 3, 4, 5$):

$$\begin{bmatrix} \frac{k_1}{s} + ms + \frac{k_2}{s} + R_1 & -\frac{k_2}{s} & 0 & 0 & 0 \\ -\frac{k_2}{s} & \frac{k_2}{s} + \frac{k_3}{s} & -\frac{k_3}{s} & 0 & 0 \\ 0 & -\frac{k_2}{s} & \frac{k_3}{s} + \frac{k_4}{s} & -\frac{k_4}{s} & 0 \\ 0 & 0 & -\frac{k_4}{s} & \frac{k_4}{s} + R_2 & -R_2 \\ 0 & 0 & 0 & -R_2 & R_2 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ V_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ F \end{bmatrix}$$

Conclusionary Remarks

Working through this assignment provided insight into how to derive the equations of motion (at least in the Laplace domain) of complex mechanical system without directly invoking Newton's Second Law. Using the electrical-mechanical analogy, circuits can adequately represent these mechanical systems, which allows circuit analysis to do the heavy lifting.

Despite each of the above mechanical systems being distinct in their own rights, reducing them to circuits pulls them down to the same level in which they can all be interpreted (almost) the same. This is analogous to how the EOMs of all these systems can be written a systems of second order linear ODEs.

As grotesque as the resultant matrix equations from these circuit diagram are, they can all be solved numerically, which trivializes these mechanical systems in totality.